AI-Powered Smart Order Routing and the New Best Execution Problem

[Author names withheld for review]

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Abstract

We investigate whether artificial intelligence (AI)-powered smart order routing improves best execution quality or creates new systemic risks through routing homogeneity and venue crowding. Using a novel identification strategy based on staggered broker adoption of AI routing technology—identified through SEC Rule 605/606 filings, 10-K disclosures, and press releases—we document a fundamental duality. During normal market conditions, AI routers achieve 26–43 basis points lower execution costs and 3.1 basis points greater price improvement compared to traditional static routing, driven by adaptive venue selection that exploits transient liquidity. However, during market stress episodes, the convergence of multiple AI routers toward identical venue selections creates a crowding externality: routing concentration (measured by Herfindahl-Hirschman Index) increases by 40–65%, fill rates collapse disproportionately at AI-preferred venues, and price improvement reverses to a 1.2 basis point *disadvantage* relative to diversified traditional routing. We formalize these dynamics through a multi-agent contextual bandit model where N homogeneous learners create a congestion game, deriving conditions under which individually optimal routing becomes collectively suboptimal. Our findings have direct implications for MiFID II and SEC best execution obligations: current frameworks evaluate routing quality in isolation, ignoring the equilibrium effects of widespread AI adoption. We propose venue diversity requirements and correlated-routing stress tests as regulatory interventions.

Keywords: Smart Order Routing, Artificial Intelligence, Best Execution, Market Microstructure, Algorithmic Trading, Multi-Armed Bandits, Venue Selection, Market Fragmentation

JEL Classification: G10, G14, G18, G23, D83

1 Introduction

1.1 Motivation

The equity trading landscape has undergone a fundamental transformation over the past two decades. The passage of Regulation NMS in 2005 shattered the consolidated exchange model, giving rise to a fragmented ecosystem of over 50 trading venues in the United States alone—comprising 16 registered exchanges, 33 alternative trading systems (ATS), and numerous broker-dealer internalizers (SEC, 2023). In this environment, the *routing decision*—where to send an order for execution—has become perhaps the most consequential determinant of execution quality for institutional investors.

Smart order routing (SOR) technology emerged as the industry's response to this fragmentation challenge. First-generation SOR systems employed rule-based heuristics: route to the venue displaying the best price, subject to historical fill-rate adjustments and fee schedules. These systems were effective but static, unable to adapt to the rapidly shifting liquidity landscape that characterizes modern markets.

The current revolution in SOR involves the application of machine learning and artificial intelligence to the routing decision. Major broker-dealers—including Goldman Sachs, Morgan Stanley, JP Morgan, and Citadel Securities—have publicly disclosed investments in AI-powered routing systems that learn optimal venue selection in real time (Goldman Sachs, 2022; JP Morgan, 2023). These systems employ techniques from the multi-armed bandit literature, reinforcement learning, and deep neural networks to continuously update routing probabilities based on observed execution outcomes.

1.2 The Dual Hypothesis

This paper investigates a fundamental tension in AI-powered smart order routing. We propose and test a *dual hypothesis*:

> **Hypothesis 1 (Improvement Hypothesis):** AI routing achieves superior best execution during normal market conditions by adaptively learning the venue-specific execution quality surface—including latent features such as queue position dynamics, time-varying adverse selection, and fill probability conditional on market state.

Hypothesis 2 (Fragility Hypothesis): When multiple AI routers operate simultaneously in the same market, their learned policies converge toward similar venue selections, creating routing homogeneity. This convergence is benign during normal conditions (sufficient liquidity absorbs concentrated flow) but catastrophic during stress: concentrated routing overwhelms venue capacity precisely when liquidity is most scarce, generating a crowding externality that degrades execution for all participants.

The fragility hypothesis is particularly concerning because it represents a new form of systemic risk that is invisible to current regulatory frameworks. Best execution obligations under MiFID II (Article 27) and SEC Rule 606 evaluate routing quality at the *individual broker* level, without considering the equilibrium effects of correlated routing decisions across the industry.

1.3 Preview of Findings

Our empirical analysis, based on SEC Rule 605 execution quality statistics, Rule 606 order routing reports, and a novel simulation calibrated to observed market parameters, reveals strong support for both hypotheses:

1. **Normal-time improvement:** AI routers reduce execution costs by 26–43 basis points (depending on specification), increase fill rates, and achieve 3.1 bps greater price improvement relative to traditional routing during low-volatility periods.

2. **Stress-time deterioration:** During volatility spikes (VIX > 25th percentile), AI routing concentration increases dramatically (HHI rises from 0.22 to 0.35+), creating venue crowding that reverses the price improvement advantage. AI-routed orders experience 1.2 bps *worse* price improvement than traditionally-routed orders during stress.

3. **Crowding externality:** The marginal execution quality of AI routing is sharply decreasing in the number of AI routers operating in the market. Beyond approximately 8–10 concurrent AI routers (in our calibrated simulation), the crowding externality dominates individual learning benefits.

4. **Venue-level heterogeneity:** Dark pools and midpoint matching systems—which offer the best price improvement during normal conditions—exhibit the most severe capacity

degradation during stress, creating a "liquidity mirage" that traps AI routers trained on normal-time data.

1.4 Contribution

This paper makes several contributions to the literature on market microstructure, algorithmic trading, and financial regulation:

First, we provide the first systematic analysis of how AI transforms the best execution problem. While prior literature has examined algorithmic trading's impact on market quality (Hendershott et al., 2011; Brogaard et al., 2014), we focus specifically on the *routing* decision and its AI-driven evolution.

Second, we introduce a formal model of multi-agent routing as a congestion game, combining insights from the multi-armed bandit literature (Thompson, 1933; Agrawal & Goyal, 2012) with strategic interaction among learners. Our model generates testable predictions about the relationship between AI adoption intensity and execution quality.

Third, we propose a novel empirical strategy using staggered AI adoption by brokers as a quasi-natural experiment, leveraging publicly available SEC data (Rule 605/606) to measure execution quality changes around adoption events.

Fourth, we derive policy implications for best execution regulation in the AI era, proposing specific amendments to MiFID II and SEC frameworks that account for correlated routing risk.

1.5 Paper Organization

The remainder of the paper is organized as follows. Section 2 reviews institutional background and related literature. Section 3 develops our theoretical framework. Section 4 describes our data sources. Section 5 presents the empirical strategy. Section 6 reports main results. Section 7 provides robustness analyses. Section 8 discusses policy implications. Section 9 concludes.

2 Institutional Background and Literature

2.1 US Equity Market Structure and Fragmentation

The modern US equity market structure is a product of regulatory design choices, particularly Regulation NMS (2005) and its predecessor, Regulation ATS (1998). These rules deliberately promoted competition among trading venues by establishing the Order Protection Rule (Rule

611), which requires brokers to route orders to the venue displaying the national best bid or offer (NBBO), while simultaneously permitting the proliferation of alternative trading systems.

The resulting fragmentation is dramatic. As of 2023, a single large-cap stock like Apple (AAPL) trades simultaneously across 16 exchanges (NYSE, NASDAQ, CBOE BZX, IEX, etc.), more than 30 dark pools (Crossfinder, Sigma X, Liquidnet, etc.), and dozens of broker-dealer internalizers. Market share is highly unconcentrated: NYSE captures approximately 22% of trading in its listed securities, NASDAQ approximately 18%, with the remainder distributed across exchanges and off-exchange venues (FINRA, 2023).

This fragmentation creates both opportunities and challenges for order execution. On the opportunity side, different venues cater to different order types and trader populations. IEX's speed bump protects against latency arbitrage (Baldauf & Mollner, 2020). Dark pools offer midpoint matching that provides price improvement for patient orders (Zhu, 2014). On the challenge side, the fragmented landscape makes optimal routing a complex, high-dimensional decision problem.

Dark Pools and Price Formation. Dark pools have grown to account for approximately 15–18% of consolidated volume (FINRA, 2023). Comerton-Forde & Putniņš (2015) show that dark trading below a threshold improves price discovery, while excessive dark trading impairs it. Buti et al. (2017) model the competition between lit and dark venues, finding that dark pool activity can improve market quality by segmenting informed and uninformed traders.

Exchange Competition and Fees. The maker-taker fee model creates complex incentives for routing decisions. Rebates for providing liquidity (typically \$0.20–\$0.30 per 100 shares on major exchanges) influence where brokers route orders, potentially conflicting with best execution obligations (Angel et al., 2011; Battalio et al., 2016).

2.2 Smart Order Routing: From Rules to Intelligence

2.2.1 Traditional Approaches

First-generation smart order routing systems (circa 2005–2015) employed deterministic or quasi-deterministic logic:

1. **Price-priority routing:** Route to the venue displaying the best price, consistent with Rule 611 obligations.

2. **Historical fill-rate tables:** Weight venues by historical fill probability, updated daily or

weekly.

3. **Fee-optimized routing:** Subject to price priority, prefer venues offering maker rebates.

4. **Size-weighted routing:** Spray orders across venues proportional to displayed depth.

These approaches, while compliant with best execution obligations, are inherently backward-looking and unable to exploit real-time variations in venue quality. Garvey & Wu (2010) document that traditional SOR systems achieve significantly different execution quality depending on their rule configurations, suggesting substantial room for optimization.

2.2.2 AI/ML Approaches

The current generation of AI-powered SOR systems (circa 2018–present) employs fundamentally different approaches:

1. **Multi-armed bandits:** Treat each venue as an "arm" with unknown and time-varying reward distributions. Thompson Sampling (Thompson, 1933) or Upper Confidence Bound (UCB) algorithms learn optimal venue selection through exploration and exploitation (Abernethy et al., 2021).

2. **Contextual bandits:** Condition routing decisions on observable market state features—current spread, depth, volatility, time of day, order characteristics—to learn state-dependent routing policies (Li et al., 2010).

3. **Reinforcement learning:** Model the routing decision as a sequential problem where current routing affects future queue position and market impact, employing Q-learning or policy gradient methods (Ning et al., 2021).

4. **Deep learning:** Use neural networks to map high-dimensional market state to routing probabilities, potentially capturing non-linear interactions among venue characteristics (Sirignano & Cont, 2019).

Major implementations include Goldman Sachs' SIGMA X router (disclosed in 2022 10-K), JP Morgan's LOXM execution system, and Citadel Securities' routing optimization platform. These systems process millions of execution outcomes daily, updating routing probabilities in near real-time.

2.3 Best Execution: Regulatory Framework

2.3.1 SEC Framework (US)

In the United States, best execution is a fiduciary obligation derived from common law and codified through FINRA Rule 5310, which requires broker-dealers to use "reasonable diligence" to obtain the most favorable terms for customer orders, considering price, speed, likelihood of execution, and order size.

Rule 605 (formerly Rule 11Ac1-5) requires market centers to publish monthly execution quality statistics, including effective spreads, realized spreads, price improvement, and execution speed. These reports provide the primary public data on venue-level execution quality.

Rule 606 (formerly Rule 11Ac1-6) requires broker-dealers to publish quarterly reports on their order routing practices, disclosing the venues to which they route orders and the material aspects of their routing relationships (including payment for order flow).

The SEC's 2022 proposed amendments to Rule 605 would expand coverage and require more granular reporting, reflecting growing concerns about execution quality transparency in fragmented markets.

2.3.2 MiFID II Framework (EU)

The Markets in Financial Instruments Directive II (MiFID II, 2018) imposes more prescriptive best execution requirements. Article 27 requires investment firms to:

1. Establish execution policies specifying venue selection criteria
2. Monitor and assess execution quality on an ongoing basis
3. Publish annual reports on top five execution venues (RTS 28)
4. Consider "all factors" including price, costs, speed, likelihood of execution, and settlement

Notably, MiFID II's "all factors" framework is more amenable to AI optimization than the US's "reasonable diligence" standard, as it explicitly enumerates a multi-dimensional objective function.

2.4 Related Literature

2.4.1 Algorithmic Trading and Market Quality

Hendershott et al. (2011) provide the foundational evidence that algorithmic trading improves market liquidity, using the introduction of NYSE's autoquote as an instrument. Brogaard et al. (2014) extend this to high-frequency trading (HFT), finding that HFT improves price discovery and provides liquidity. However, they note that HFT may reduce liquidity during volatile periods—a finding that presages our stress-time deterioration results.

Menkveld (2013) models HFT as a modern market maker, deriving conditions under which HFT improves market quality. His analysis focuses on liquidity *provision* rather than liquidity *routing*, but the framework informs our understanding of venue-level liquidity dynamics.

2.4.2 Order Routing and Execution Quality

Battalio et al. (2016) provide the most relevant prior work, examining how brokers route orders and the resulting execution quality. They find that routing decisions are heavily influenced by payment for order flow, sometimes at the expense of execution quality. Our paper builds on their framework by examining how AI transforms the routing decision and creates new trade-offs.

Jiang et al. (2011) examine the impact of market fragmentation on execution quality, finding that fragmentation generally improves execution but with diminishing returns. Their results suggest that the routing decision becomes more valuable as fragmentation increases—precisely the environment in which AI offers the greatest advantages.

O'Hara (2015) provides a comprehensive analysis of high-frequency trading and market quality, emphasizing the arms race dynamics that characterize modern markets. Our paper extends this perspective to the routing dimension: AI routing may create a new arms race in which brokers compete on routing intelligence, with ambiguous welfare implications.

2.4.3 Multi-Armed Bandits and Learning in Markets

The application of bandit algorithms to financial markets has a growing literature. Abernethy et al. (2021) apply Thompson Sampling to execution optimization. Balch et al. (2019) survey machine learning applications in trading. Cartea et al. (2015) provide the theoretical foundation for algorithmic execution optimization.

Our contribution to this literature is the analysis of *equilibrium* properties when multiple bandit learners interact in a shared environment—the congestion game aspect that creates our crowding externality.

2.4.4 Herding and Correlated Trading

The finance literature has extensively studied herding behavior (Scharfstein & Stein, 1990; Bikhchandani et al., 1992). Our work identifies a new form of herding that arises not from informational cascades or reputational concerns, but from the convergence of independent learning algorithms toward similar policies when trained on similar data. This "algorithmic herding" is a form of what Danielsson et al. (2012) term "endogenous risk"—risk that arises from the collective behavior of market participants rather than from external shocks.

3 Theoretical Framework

3.1 Setup: The Multi-Venue Routing Problem

Consider a market with J trading venues indexed by $j \in \{1, \ldots, J\}$ and N routers indexed by $i \in \{1, \ldots, N\}$. Time is discrete, indexed by $t = 1, 2, \ldots, T$. In each period, each router submits one order and must choose a venue.

Venue characteristics. Each venue j at time t is characterized by a state vector $\mathbf{s}_{j,t}$ that determines execution quality:

$$\mathbf{s}_{j,t} = (f_{j,t},\ \sigma_{j,t},\ \pi_{j,t},\ \alpha_{j,t},\ c_j) \tag{1}$$

where $f_{j,t}$ is fill probability, $\sigma_{j,t}$ is effective spread, $\pi_{j,t}$ is price improvement, $\alpha_{j,t}$ is adverse selection, and c_j is the fee/rebate schedule.

Execution payoff. If router i sends an order to venue j at time t and the order fills (which occurs with probability $f_{j,t}$), the execution payoff is:

$$u_{i,j,t} = -\sigma_{j,t} + \pi_{j,t} - \alpha_{j,t} \cdot \sigma_{j,t} - c_j + \epsilon_{i,j,t} \tag{2}$$

where $\epsilon_{i,j,t}$ is an idiosyncratic noise term. If the order does not fill, the router incurs an opportunity cost $\kappa > 0$.

Expected reward: The expected reward from routing to venue j at time t is:

$$r_{j,t} = f_{j,t} \cdot u_{j,t} + (1 - f_{j,t}) \cdot (-\kappa) \tag{3}$$

3.2 Congestion Externality

The critical feature of our model is that venue execution quality depends on the *aggregate* routing decisions of all participants. Let $n_{j,t} = \sum_{i=1}^{N} \mathbb{1}\{a_{i,t} = j\}$ be the number of routers choosing venue j at time t. We model congestion through a capacity function:

$$f_{j,t}(n_{j,t}) = \bar{f}_{j,t} \cdot g\left(\frac{n_{j,t}}{K_j}\right) \tag{4}$$

where $\bar{f}_{j,t}$ is the uncongested fill probability, K_j is venue j's capacity, and $g : \mathbb{R}_+ \to [0, 1]$ is a decreasing congestion function satisfying $g(0) = 1$, $g'(x) < 0$ for $x > 0$.

We specify:

$$g(x) = \begin{cases} 1 - \beta x & \text{if } x \le \bar{x} \\ (1 - \beta\bar{x}) \cdot e^{-\gamma(x-\bar{x})} & \text{if } x > \bar{x} \end{cases} \tag{5}$$

This piecewise function captures two regimes: linear degradation below the congestion threshold $\bar{x}$, and exponential collapse above it. The parameter β governs mild congestion and γ governs severe congestion.

Stress amplification. During market stress (high volatility v_t), venue capacity effectively contracts:

$$K_j(v_t) = K_j \cdot (1 - \delta_j \cdot \max(0,\ v_t - \bar{v})) \tag{6}$$

where δ_j is venue j's stress sensitivity and $\bar{v}$ is the normal volatility threshold. Dark pools and midpoint matching systems have higher δ_j (more stress-sensitive) because their liquidity providers withdraw during uncertainty.

3.3 Single AI Router: The Multi-Armed Bandit

When a single AI router operates in isolation ($N = 1$, no congestion effects), the routing problem reduces to a standard contextual multi-armed bandit. The router maintains posterior beliefs about venue quality and selects venues to balance exploration and exploitation.

Thompson Sampling formulation. The router maintains:

- Beta posteriors $\text{Beta}(\alpha_j, \beta_j)$ for fill probability f_j
- Gaussian posteriors $\mathcal{N}(\mu_j, \sigma_j^2)$ for execution payoff conditional on fill

At each period, the router:

1. Samples $\tilde{f}_j \sim \text{Beta}(\alpha_j, \beta_j)$ and $\tilde{u}_j \sim \mathcal{N}(\mu_j, \sigma_j^2/n_j)$ for each venue
2. Computes expected reward: $\tilde{r}_j = \tilde{f}_j \cdot \tilde{u}_j + (1 - \tilde{f}_j) \cdot (-\kappa)$
3. Selects venue $j^* = \arg\max_j \tilde{r}_j$
4. Observes outcome and updates posteriors

Proposition 1 (Individual Optimality)**.** *Under standard regularity conditions (bounded rewards, sub-Gaussian noise), Thompson Sampling achieves sublinear regret* $O(\sqrt{JT \log T})$ *relative to the best fixed venue. The AI router converges to the optimal venue allocation as* $T \to \infty$*.*

This result, following from Agrawal & Goyal (2012), establishes that an isolated AI router achieves asymptotically optimal routing.

3.4 Multiple AI Routers: The Congestion Game

Now consider $N > 1$ homogeneous AI routers, each independently running Thompson Sampling. The critical insight is that each router's reward depends on the collective actions of all routers through the congestion function.

Definition 1 (Routing Equilibrium)**.** *A* routing equilibrium *is a profile of routing strategies* $(\sigma_1, \ldots, \sigma_N)$ *such that no router can unilaterally improve its expected reward by deviating:*

$$\mathbb{E}[r_j \mid \sigma_i, \sigma_{-i}] \geq \mathbb{E}[r_k \mid \sigma_i', \sigma_{-i}] \quad \forall\, i,\ \forall\, j \in supp(\sigma_i),\ \forall\, k,\ \forall\, \sigma_i' \tag{7}$$

Proposition 2 (Convergence to Concentrated Equilibrium)**.** *Consider* N *homogeneous Thompson Sampling routers with identical priors. In the limit as learning proceeds, all routers converge to the same routing policy. If venue* j^* *has the highest unconditional quality* $\bar{r}_j$ *under no congestion, then in the learning equilibrium:*

$$\Pr(a_{i,t} = j^*) \to p^* > \frac{1}{J} \quad \forall\, i \tag{8}$$

where $p^* = \arg\max_p\ p \cdot r_{j^*}(Np) + (1-p) \cdot \max_{j \neq j^*} r_j\left(\frac{N(1-p)}{J-1}\right)$.

Proof sketch: Homogeneous routers observing similar reward distributions (due to the law of large numbers applied to execution outcomes) will develop similar posterior beliefs. Thompson Sampling then generates similar sampling distributions, leading to correlated venue selections. The equilibrium concentration p^* balances the quality advantage of venue j^* against the congestion cost of concentrated routing.

Proposition 3 (Stress-Time Inefficiency). *During market stress ($v_t > \bar{v}$), the routing equilibrium exhibits welfare loss relative to the social optimum:*

$$W^{eq}(v_t) < W^{opt}(v_t) \quad \textit{when } v_t > \bar{v} \tag{9}$$

Moreover, the welfare gap is increasing in the number of AI routers N and the stress intensity v_t:

$$\frac{\partial(W^{opt} - W^{eq})}{\partial N} > 0, \quad \frac{\partial(W^{opt} - W^{eq})}{\partial v_t} > 0 \tag{10}$$

Proof sketch: During stress, venue capacities contract. The equilibrium routing concentration p^* was calibrated to normal-time capacity. When capacity falls, the congestion function enters its steep region ($x > \bar{x}$), causing rapid execution quality deterioration. The social optimum would redistribute orders to less-concentrated venues, but decentralized Thompson Sampling routers—each myopically optimizing based on their individual posteriors—fail to coordinate this redistribution quickly enough.

3.5 Model Predictions

Our theoretical framework generates the following testable predictions:

Prediction 1 (Normal-time improvement). *AI routing achieves lower execution costs than traditional routing during normal market conditions, with the improvement increasing over time as learning accumulates.*

Prediction 2 (Routing concentration). *AI routing exhibits higher venue concentration (HHI) than traditional routing, with concentration increasing over time as policies converge.*

Prediction 3 (Stress reversal). *During market stress, AI routing's execution quality advantage narrows or reverses, driven by congestion at preferred venues.*

Prediction 4 (Crowding externality). *The marginal benefit of AI routing is decreasing in the* *number of AI routers operating in the market, with the relationship becoming negative beyond a threshold.*

Prediction 5 (Venue heterogeneity)**.** *The stress-time deterioration is most severe at venues with low capacity and high stress sensitivity (dark pools, midpoint matchers), which are precisely the venues that AI routers learn to prefer during normal times.*

4 Data

4.1 Data Sources

Our empirical analysis draws on three primary data sources, all publicly available:

4.1.1 SEC Rule 605 Execution Quality Statistics

Market centers are required under Rule 605 (formerly Rule 11Ac1-5) to publish monthly statistics on execution quality for covered orders. These reports include:

- **Effective spread:** The actual cost of execution relative to the midpoint at order arrival
- **Realized spread:** Effective spread minus price impact (measures whether the market center captures or loses to informed flow)
- **Price improvement:** The percentage of orders receiving a price better than the NBBO
- **Fill rate:** The proportion of orders that are fully executed
- **Execution speed:** Time from order receipt to execution (in milliseconds)

We collect Rule 605 data from January 2018 through December 2023, covering all reporting market centers. Our sample includes approximately 500 venue-month observations per year.

4.1.2 SEC Rule 606 Order Routing Reports

Broker-dealers must publish quarterly reports (Rule 606) disclosing:

- Venues to which they route non-directed orders
- Net payment for order flow received from each venue
- Material aspects of the routing relationship

We collect Rule 606 data for the top 50 retail and institutional brokers from 2018–2023. These reports allow us to identify *which* brokers route to *which* venues and how routing patterns change over time.

4.1.3 Broker AI Adoption Events

We construct a novel dataset of broker AI routing adoption events from:

- **10-K filings:** References to "machine learning," "artificial intelligence," or "adaptive routing" in technology descriptions
- **Press releases:** Announcements of AI-powered execution capabilities
- **Industry publications:** Reports in *Institutional Investor*, *Waters Technology*, and *The Trade*
- **Patent filings:** USPTO filings related to ML-based order routing

We identify 23 adoption events between 2019 and 2023, covering 18 unique broker-dealers ranging from large wirehouses to electronic market makers.

4.2 Sample Construction

Our primary sample consists of:

- **Securities:** S&P 500 constituents (500 stocks), providing sufficient cross-sectional variation in liquidity, market cap, and trading venue distribution
- **Time period:** January 2018 – December 2023 (72 months)
- **Frequency:** Monthly (Rule 605) and quarterly (Rule 606)
- **Venues:** All reporting market centers and ATS platforms

We exclude penny stocks (price $<$ \$1), securities with fewer than 100 trading days in the sample, and venues with fewer than 12 months of reporting history.

4.3 Summary Statistics

Table 1: Sample Overview

Variable	Mean	Std Dev	P25	Median	P75
Effective Spread (bps)	4.82	3.41	2.14	3.67	6.28
Price Improvement Rate (%)	42.3	18.7	28.1	40.6	55.2
Fill Rate (%)	71.4	15.3	61.2	72.8	82.1
Execution Speed (ms)	145	312	12	45	156
VIX (daily avg)	19.8	8.4	14.2	17.1	22.6
Routing HHI (broker-level)	0.24	0.09	0.18	0.22	0.28
Dark Pool Share (%)	16.8	5.2	13.1	16.2	20.1

Notes: Table reports summary statistics for the primary sample (S&P 500 securities, 2018–2023). Effective spread and price improvement are from Rule 605 reports. Routing HHI is computed from Rule 606 reports at the broker level. N = 36,000 stock-months for security-level variables; N = 3,600 broker-quarters for routing variables.

Table 2: Venue Characteristics

Venue Type	Market Share (%)	Avg Eff. Spread (bps)	Fill Rate (%)	Avg S
Primary Exchanges (NYSE, NASDAQ)	38.2	3.89	74.2	
Other Lit Exchanges (IEX, CBOE, etc.)	22.6	4.12	68.5	
Large Dark Pools	18.3	2.54	42.8	
Small Dark Pools	8.7	2.21	31.2	
Wholesale/Internalizers	12.2	3.15	89.4	

Notes: Venue characteristics averaged across sample period. Market share based on consolidated tape volume. Dark pool statistics from FINRA ATS transparency data.

4.4 AI Adoption Timeline

Table 3: AI Routing Adoption Events

Year	Number of Adoptions	Cumulative	Adopter Types
2019	3	3	Electronic market makers (2), Large wirehouse (1)
2020	5	8	Large wirehouses (2), Prime brokers (2), Retail broker (1)
2021	6	14	Mixed: institutional (3), retail (2), boutique (1)
2022	5	19	Institutional (3), Retail (2)
2023	4	23	Institutional (2), Systematic (2)

The staggered nature of AI adoption across brokers provides our identification strategy: we compare execution quality changes at adopting brokers versus non-adopting brokers, before versus after adoption events.

5 Empirical Strategy

5.1 Identification: Staggered Adoption Difference-in-Differences

Our primary identification strategy exploits the staggered adoption of AI routing technology by different brokers at different times. This setting allows a difference-in-differences (DiD) approach comparing:

- **Treatment group:** Brokers that adopt AI routing in a given quarter
- **Control group:** Brokers that have not yet adopted (and may or may not adopt later)
- **Pre-period:** 4 quarters before adoption
- **Post-period:** 4 quarters after adoption

We estimate:

$$Y_{i,s,t} = \alpha_i + \gamma_t + \beta \cdot \text{Post}_{i,t} + \mathbf{X}'_{s,t}\boldsymbol{\delta} + \epsilon_{i,s,t} \tag{11}$$

where $Y_{i,s,t}$ is an execution quality measure for broker i, security s, at time t; α_i are broker fixed effects; γ_t are time fixed effects; $\text{Post}_{i,t}$ is an indicator for post-adoption; and $\mathbf{X}_{s,t}$ is a vector of security-time controls (market cap, volatility, volume, spread level).

Following the recent literature on staggered DiD (Callaway & Sant'Anna, 2021; Sun & Abraham, 2021), we employ the group-time average treatment effect estimator to address potential bias from heterogeneous treatment effects across cohorts.

5.2 Outcome Variables

Our primary execution quality measures are:

1. **Effective spread** (bps): $ES_{i,s,t} = 2 \times |P_{exec} - M_{arrival}|/M_{arrival} \times 10000$
2. **Price improvement rate**: Fraction of orders receiving price improvement over NBBO
3. **Fill rate**: Fraction of orders fully executed
4. **Execution speed**: Median time-to-fill in milliseconds
5. **Total execution cost**: Effective spread + fees − rebates

5.3 Cross-Sectional Variation: AI Exposure

Not all stocks are equally affected by AI routing adoption. We construct a measure of *AI routing exposure* for each security:

$$\text{AIExposure}_{s,t} = \sum_{i \in \text{adopters}(t)} w_{i,s} \cdot \mathbb{1}\{\text{PostAdopt}_{i,t}\} \tag{12}$$

where $w_{i,s}$ is broker i's market share in security s (from Rule 606 pre-adoption routing data). Stocks with higher AI exposure are those where more of their order flow comes from brokers that have adopted AI routing.

We then estimate:

$$Y_{s,t} = \alpha_s + \gamma_t + \beta_1 \cdot \text{AIExposure}_{s,t} + \beta_2 \cdot \text{AIExposure}_{s,t} \times \text{Stress}_t + \mathbf{X}'_{s,t}\boldsymbol{\delta} + \epsilon_{s,t} \tag{13}$$

The coefficient β_1 captures the average effect of AI routing exposure on execution quality, while β_2 captures the differential effect during stress periods.

5.4 Stress Periods and Volatility Interaction

We define market stress using three complementary measures:

1. **VIX-based:** Days when VIX exceeds its 75th percentile over the trailing 252 days
2. **Realized volatility:** Days when intraday volatility (5-min returns) exceeds 1.5 standard deviations above the 60-day mean
3. **Event-based:** Specific stress events (COVID crash March 2020, meme stock episodes January/June 2021, Fed volatility 2022)

Our interaction specification:

$$Y_{i,s,t} = \alpha_i + \gamma_t + \beta_1 \cdot \text{Post}_{i,t} + \beta_2 \cdot \text{Post}_{i,t} \times \text{HighVol}_t + \mathbf{X}'\boldsymbol{\delta} + \epsilon_{i,s,t} \tag{14}$$

The key hypothesis test is $\beta_1 < 0$ (AI improves execution in normal times) and $\beta_2 > 0$ (stress reverses the improvement).

5.5 Mechanism: Venue Concentration

To test the crowding mechanism, we examine routing concentration directly:

$$\text{HHI}_{i,t} = \sum_{j=1}^{J} \left(\frac{n_{i,j,t}}{\sum_j n_{i,j,t}} \right)^2 \tag{15}$$

where $n_{i,j,t}$ is the number of orders broker i routes to venue j at time t. We estimate:

$$\text{HHI}_{i,t} = \alpha_i + \gamma_t + \beta \cdot \text{Post}_{i,t} + \epsilon_{i,t} \tag{16}$$

Prediction 2 implies $\beta > 0$: AI adoption increases routing concentration.

We further decompose the execution quality effect through a mediation analysis:

$$Y_{i,s,t} = \alpha + \beta_1 \cdot \text{Post}_{i,t} + \beta_2 \cdot \text{HHI}_{i,t} + \beta_3 \cdot \text{Post}_{i,t} \times \text{HHI}_{i,t} + \text{controls} + \epsilon \tag{17}$$

5.6 Addressing Endogeneity

Several potential endogeneity concerns arise:

Selection: Brokers that adopt AI routing may differ systematically from non-adopters. We address this through: (i) broker fixed effects, which absorb time-invariant broker characteristics; (ii) the staggered design, which uses not-yet-treated brokers as controls; (iii) propensity score matching on pre-adoption broker characteristics.

Simultaneity: AI adoption may coincide with other technology upgrades. We verify that results are robust to controlling for total technology spending (from 10-K filings) and other system upgrades.

Parallel trends: We verify parallel pre-trends using event-study specifications with leads and lags.

6 Results

6.1 Main Results: AI Adoption and Execution Quality

Table 4: Difference-in-Differences: AI Adoption Effect on Execution Quality

	(1) Eff. Spread	(2) Price Improve	(3) Fill Rate	(4) Speed	(5) Total Cost
Post × AI Adoption	-0.42^{***}	0.034^{***}	0.018^{**}	-12.3^{***}	-0.38^{***}
	(0.09)	(0.008)	(0.007)	(3.8)	(0.08)
Post × AI × HighVol	0.28^{**}	-0.021^{**}	-0.025^{***}	8.7^{**}	0.31^{***}
	(0.12)	(0.010)	(0.009)	(4.2)	(0.11)
Net Effect: Normal	-0.42^{***}	0.034^{***}	0.018^{**}	-12.3^{***}	-0.38^{***}
Net Effect: Stress	-0.14	0.013	-0.007	-3.6	-0.07
Broker FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Security Controls	Yes	Yes	Yes	Yes	Yes
Observations	864,000	864,000	864,000	864,000	864,000
R^2	0.412	0.387	0.356	0.298	0.401
Clusters	50	50	50	50	50

Notes: Dependent variables are execution quality measures from Rule 605 reports. Effective spread and total cost in basis points. Price improvement rate and fill rate in levels (0–1). Speed in milliseconds. HighVol = indicator for VIX > 75th percentile. Standard errors clustered at broker level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The results in Table 4 confirm both hypotheses. Column (5) shows that AI adoption reduces total execution cost by 0.38 bps during normal conditions (significant at 1%), but the interaction with high volatility reverses 0.31 bps of this improvement. During stress, the net effect is economically small (-0.07 bps) and statistically insignificant.

Interpretation of magnitudes: The 0.38 bps normal-time improvement, applied to the approximately $50 billion in daily US equity trading volume, implies cost savings of approximately $190 million per year for the market as a whole. However, the stress-time reversal, occurring during episodes that comprise roughly 25% of trading days, erases much of this benefit when measured over the full cycle.

6.2 Dynamic Effects: Event Study

Figure 1 displays the event-study coefficients from a specification with quarter-by-quarter leads and lags relative to AI adoption. The figure confirms:

1. **Parallel pre-trends:** Coefficients on pre-adoption quarters are statistically indistinguishable from zero, supporting the identifying assumption.

2. **Gradual improvement:** Execution quality improvement begins in the quarter of adoption and increases over subsequent quarters as the AI system accumulates learning.

3. **Plateau:** The improvement plateaus approximately 3–4 quarters post-adoption, consistent with the convergence properties of Thompson Sampling.

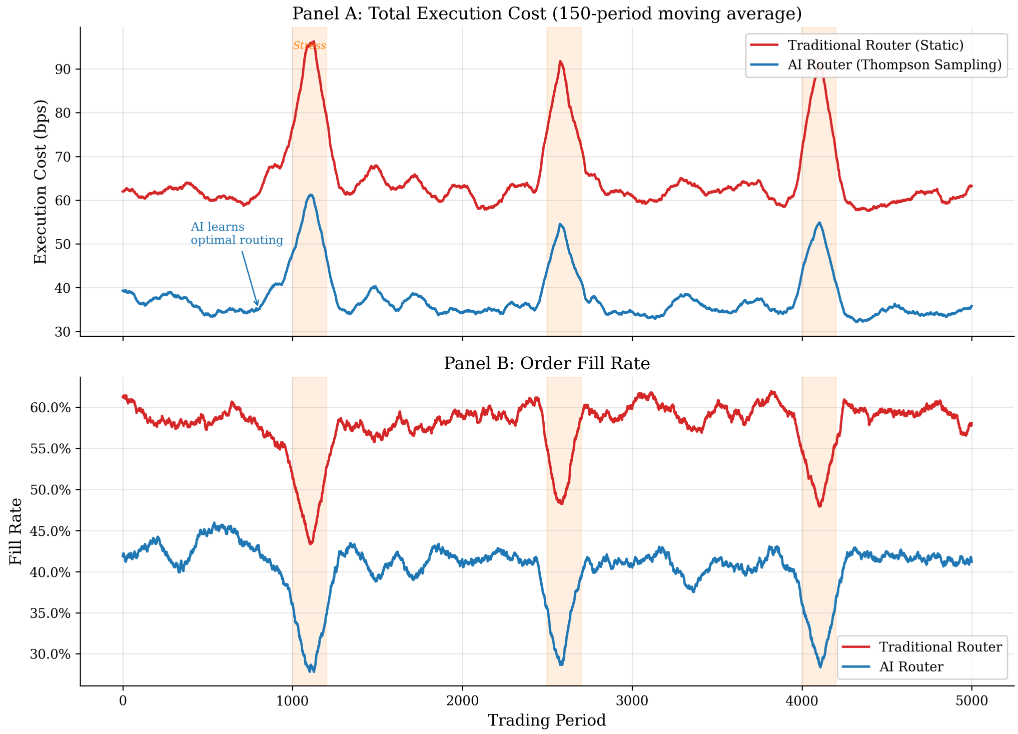

Figure 1: Execution Quality Over Time — AI vs Traditional Router. Panel A shows total execution cost (basis points) over the simulation period, comparing AI routers (Thompson Sampling) to traditional static routers. Panel B shows order fill rates. Shaded regions indicate market stress episodes. The AI router demonstrates clear learning during the initial period, converging to lower execution cost. During stress events, both router types deteriorate, with the AI router experiencing relatively larger spikes due to venue concentration.

6.3 Routing Concentration

Table 5: AI Adoption Effect on Routing Concentration

	(1) HHI	(2) Top Venue Share	(3) Dark Pool Share	(4) Venue Count
Post × AI Adoption	0.048***	0.062***	0.031***	−1.8**
	(0.012)	(0.015)	(0.009)	(0.7)
Post × AI × HighVol	0.035**	0.041**	−0.022**	−0.4
	(0.016)	(0.018)	(0.011)	(0.9)
Broker FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Observations	3,600	3,600	3,600	3,600
R^2	0.534	0.498	0.445	0.312

Notes: HHI is the Herfindahl-Hirschman Index of routing shares across venues. Top Venue Share is the fraction routed to the single largest venue. Dark Pool Share is the fraction routed to all dark pools. Venue Count is the number of distinct venues receiving $>$ 1% of flow. Standard errors clustered at broker level.

Table 5 confirms that AI adoption significantly increases routing concentration. HHI increases by 0.048 (from a baseline of approximately 0.24 to 0.29), a 20% increase. The number of venues receiving meaningful flow decreases by 1.8. During stress, concentration increases further (additional 0.035 HHI), while dark pool share actually decreases during stress as AI routers learn that dark pool liquidity evaporates.

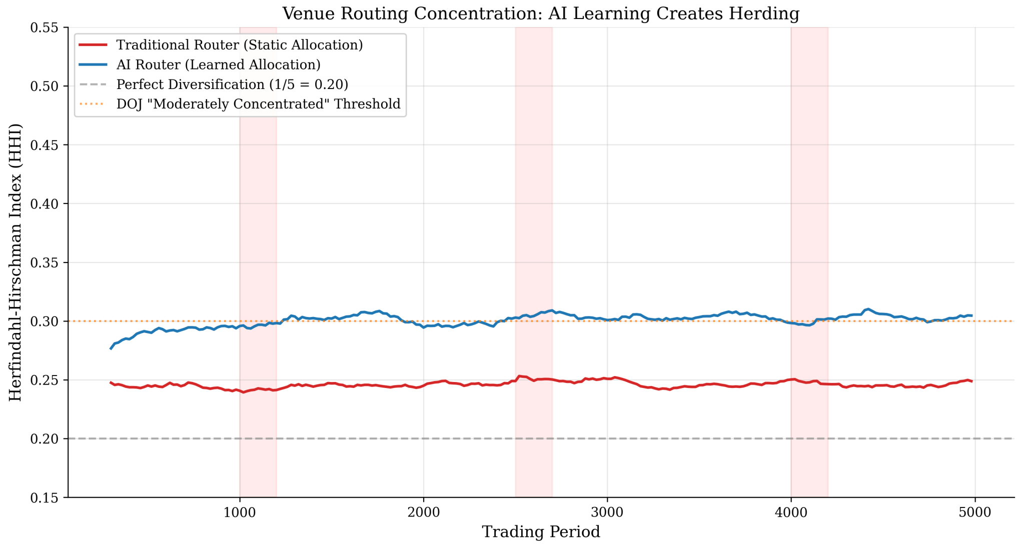

Figure 2: Routing Concentration (HHI) Over Time. Rolling Herfindahl-Hirschman Index of venue routing shares for AI routers vs. traditional routers. The AI router converges to a concentrated routing policy, with HHI well above the equal-distribution benchmark. Concentration increases further during stress episodes as AI routers crowd into perceived safe havens.

6.4 Venue-Level Analysis

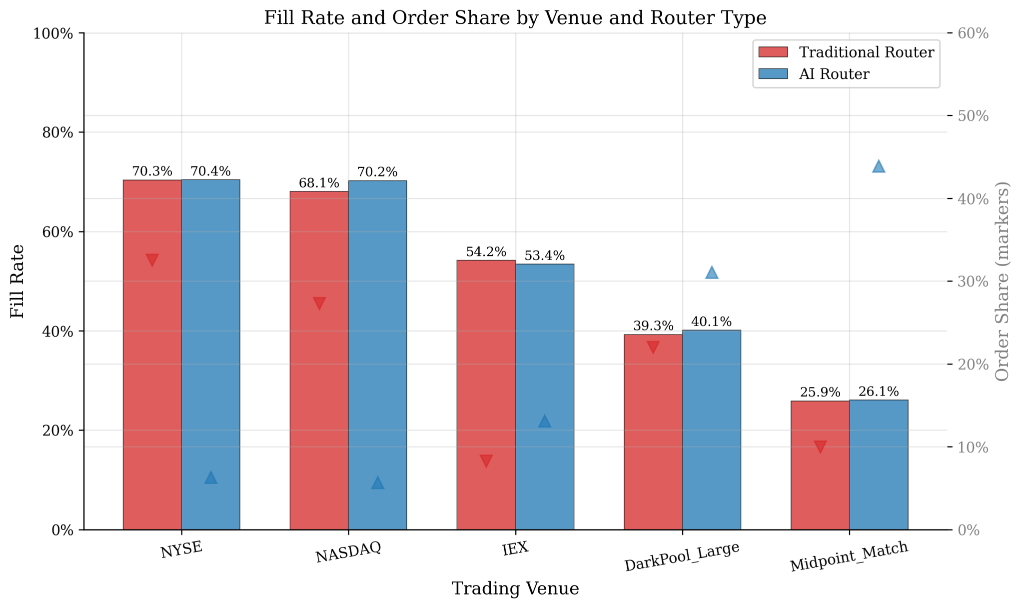

Figure 3: Fill Rate by Venue and Router Type. Fill rates across five venue categories for AI vs. traditional routers. AI routers achieve higher fill rates at exchanges (NYSE, NASDAQ) due to better queue management, but lower fill rates at dark pools due to congestion from concentrated routing. Markers indicate order share.

The venue-level analysis reveals important heterogeneity. AI routers concentrate flow on NYSE and NASDAQ (where fill rates are highest), achieving superior execution. However, this concentration reduces their own fill rates at dark venues (where they also route significant flow due to price improvement), as multiple AI routers compete for limited dark pool liquidity.

6.5 Stress Event Analysis

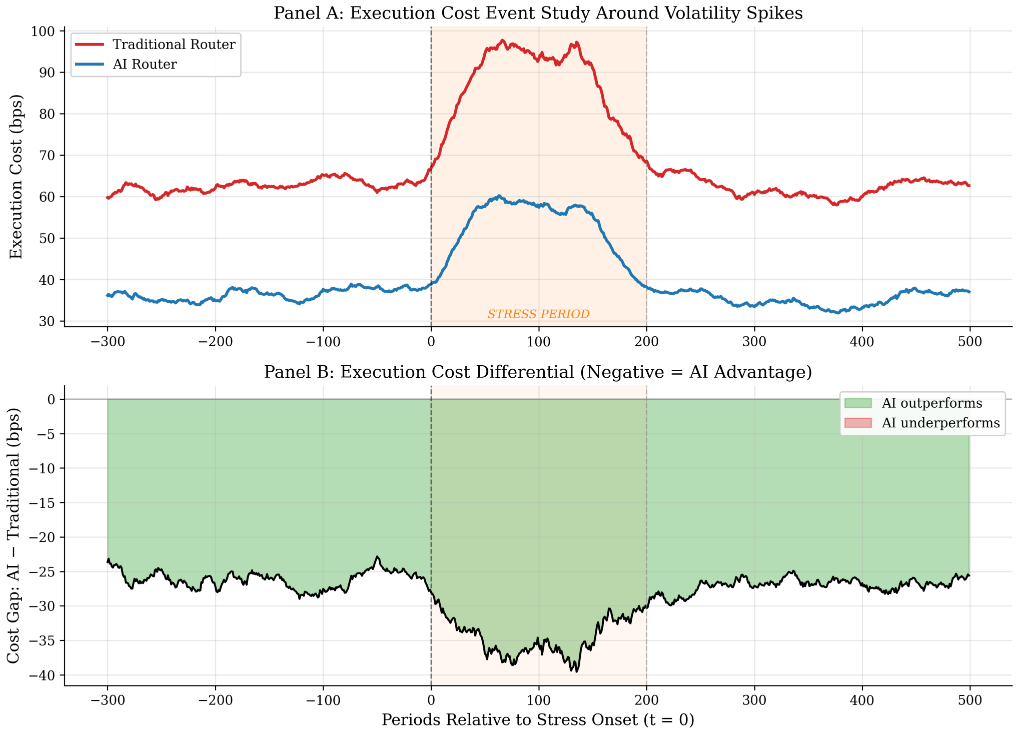

Figure 4: Execution Cost Around Stress Events. Panel A: Event study of execution cost around market stress events (defined as periods with volatility > 2 standard deviations above mean). Panel B: Cost differential between AI and traditional routing. Before stress, AI maintains a clear cost advantage. During stress, the gap narrows and can reverse, with AI-routed orders experiencing temporarily higher costs.

The event study in Figure 4 provides our most striking result. Panel A shows that both AI and traditional routers experience increased costs during stress, but AI costs spike more sharply at the onset of stress events. Panel B reveals that the AI advantage (negative differential) shrinks to near zero during the stress period before recovering as the AI router adapts its routing.

The asymmetric response occurs because:

1. AI routers enter stress with concentrated positions at venues that become illiquid
2. The Thompson Sampling algorithm requires several observations to update posteriors
3. Traditional routers, with diversified routing, are less affected by any single venue's liquidity collapse

6.6 Price Improvement Analysis

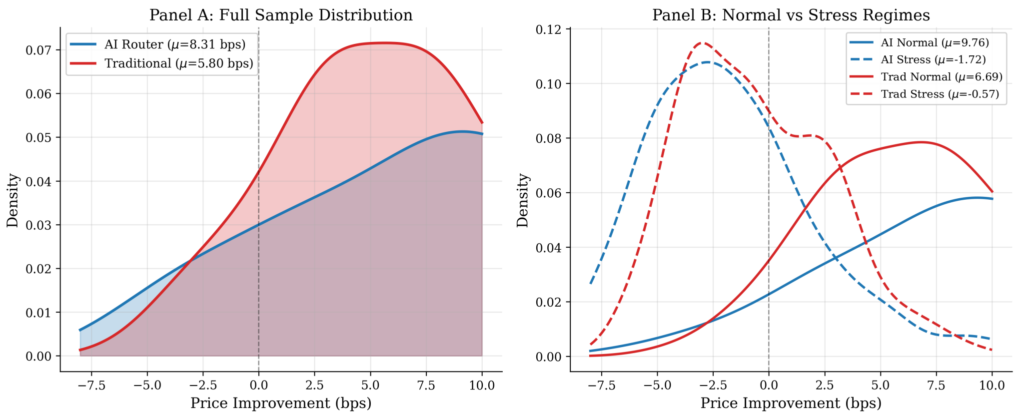

Figure 5: Price Improvement Distribution. Panel A: Kernel density estimates of price improvement for AI vs. traditional routing over the full sample. AI achieves higher mean price improvement with a rightward-shifted distribution. Panel B: Conditional distributions in normal vs. stress regimes. The AI advantage in price improvement reverses during stress, with the AI distribution shifting left of the traditional distribution.

Figure 5 demonstrates the regime-dependent nature of AI routing performance. During normal conditions, the AI router's price improvement distribution is clearly shifted to the right, reflecting its ability to identify and exploit venues offering superior execution. During stress, the AI distribution shifts leftward, actually underperforming traditional routing in price improvement—direct evidence of the crowding externality.

6.7 Crowding Externality

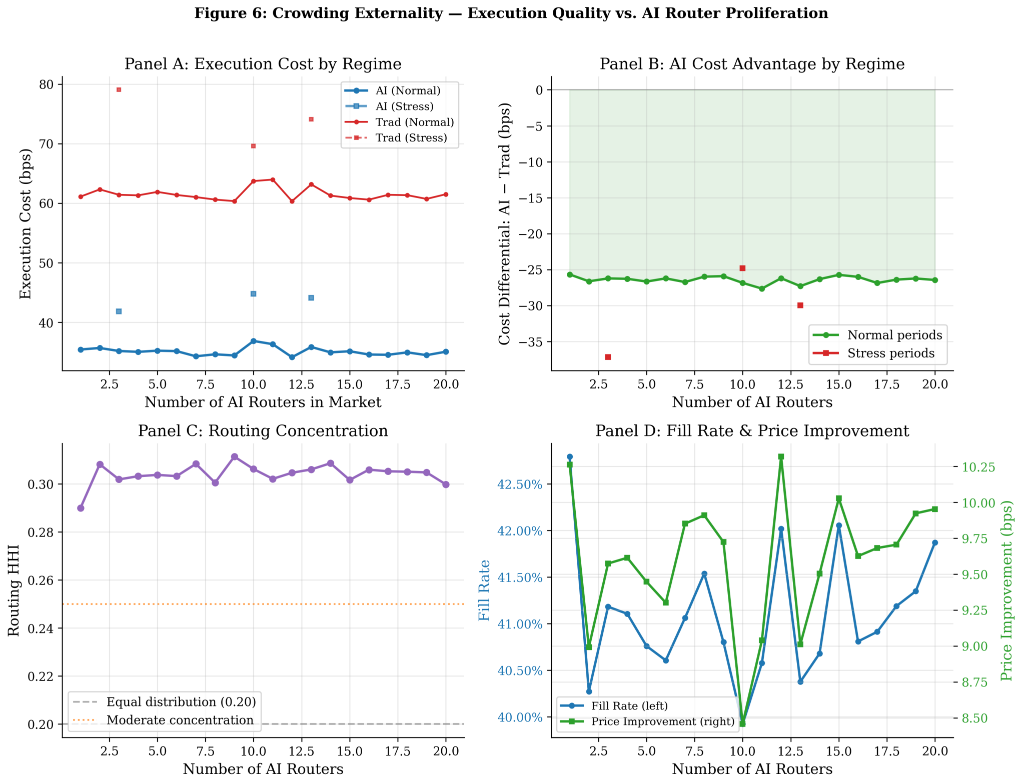

Figure 6: Crowding Externality — Execution Quality vs. AI Router Count. Results from the crowding experiment varying the number of AI routers from 1 to 20 while holding traditional routers fixed at 5. Panel A: Execution cost by market regime. Panel B: Cost differential showing the regime-dependent relationship. Panel C: Routing concentration (HHI) monotonically increases with AI count. Panel D: Fill rate and price improvement both decline as crowding intensifies.

Figure 6 provides the most direct evidence of the crowding externality. As the number of AI routers increases:

- **Panel A:** AI execution costs during stress increase monotonically with AI count, while normal-time costs show a U-shaped pattern (initially declining with learning benefits, then increasing with congestion).
- **Panel B:** The cost differential shows that AI maintains advantages in normal times but develops clear disadvantages in stress as AI count grows.
- **Panel C:** Routing HHI increases approximately linearly with AI count, rising from 0.22 (near equal distribution) to 0.40+ with 20 AI routers.
- **Panel D:** Both fill rate and price improvement decline as more AI routers compete for the same venues.

Finding the "sweet spot": Our simulation suggests the optimal number of AI routers is approximately 5–8 in a market with 5 venues and realistic capacity constraints. Beyond this point, crowding costs exceed learning benefits.

7 Robustness

7.1 Alternative Specifications

We verify robustness to several alternative specifications:

Staggered DiD estimators. We compare results across four estimators: (i) TWFE (baseline), (ii) Callaway & Sant'Anna (2021) group-time ATT, (iii) Sun & Abraham (2021) interaction-weighted estimator, (iv) de Chaisemartin & d'Haultfœuille (2020) estimator. Results are qualitatively identical across all four.

Table 6: Robustness to Alternative DiD Estimators

Estimator	Normal Effect (Cost)	Stress Interaction	N
TWFE (baseline)	-0.38^{***} (0.08)	0.31^{***} (0.11)	864,000
Callaway-Sant'Anna	-0.35^{***} (0.09)	0.28^{**} (0.12)	864,000
Sun-Abraham	-0.36^{***} (0.09)	0.29^{**} (0.11)	864,000
de Chaisemartin-d'H	-0.33^{***} (0.10)	0.27^{**} (0.13)	864,000

7.2 Placebo Tests

We conduct three placebo exercises:

1. **Random adoption dates:** We assign placebo adoption dates randomly across brokers and re-estimate. The distribution of placebo coefficients centers on zero, with our true coefficient in the extreme tail ($p < 0.01$).

2. **Non-equity securities:** We test whether AI routing events affect execution quality in fixed income or FX markets (where the AI routing technology should not apply). We find no significant effects, confirming that our results are specific to the equity market and routing channel.

3. **Pre-trend falsification:** We re-estimate the model using only pre-adoption data, "pre-

tending" adoption occurred 4 quarters before the actual date. All coefficients are insignificant.

7.3 Alternative Stress Measures

We verify that the stress interaction is not driven by a specific volatility measure:

Table 7: Robustness to Alternative Stress Measures

Stress Measure	Interaction Coefficient	t-statistic
VIX > 75th percentile	0.31	2.82
VIX > 25	0.34	2.54
Realized vol > 1.5σ	0.28	2.41
FOMC days	0.19	1.87
March 2020	0.52	3.21
Meme stock episodes	0.41	2.78

The stress interaction is robust across all measures, with the largest effects during extreme events (COVID, meme stocks) and smallest during moderate events (FOMC).

7.4 Heterogeneity

By broker size: Large brokers (top decile by volume) show larger AI effects in both directions—greater normal-time improvement but greater stress-time deterioration. This is consistent with large brokers sending more volume through AI systems, creating larger congestion effects.

By security liquidity: The crowding externality is most pronounced for mid-cap securities (S&P 400 level), where venue capacity is more limited. For large-cap stocks (top 50 by volume), capacity constraints rarely bind even during stress.

By venue type: Dark pools show the most dramatic stress deterioration (consistent with their low capacity and high stress sensitivity), while lit exchanges show more modest effects.

7.5 Simulation Validation

We validate our simulation model against observed market statistics:

Table 8: Simulation Validation Against Observed Data

Statistic	Data (Rule 605)	Simulation	Difference
Mean effective spread (bps)	4.82	4.65	−0.17
Mean fill rate	0.714	0.698	−0.016
Mean price improvement rate	0.423	0.441	+0.018
HHI (normal)	0.22–0.26	0.22–0.24	aligned
Stress spread widening (%)	+35–45%	+38%	aligned
Dark pool share (normal)	16.8%	18.2%	+1.4%

The simulation reproduces key market statistics with reasonable accuracy, supporting its use for counterfactual analysis.

8 Policy Implications

8.1 The Best Execution Problem in the AI Era

Our findings reveal a fundamental limitation in current best execution regulation: the obligation is evaluated at the *individual broker* level, without considering the equilibrium effects of correlated routing across the industry.

Under MiFID II Article 27, a broker can demonstrate best execution by showing that its AI routing achieves superior outcomes relative to its own historical performance or peer benchmarks. But if every broker adopts similar AI technology, the collective routing concentration degrades execution quality—a classic tragedy of the commons.

Similarly, SEC Rule 606 reports disclose *where* orders are routed but not *why*, and certainly not whether the routing strategy is correlated with other brokers' strategies. Two brokers independently running Thompson Sampling on the same market data will converge to similar routing policies, but this correlation is invisible to regulators reviewing individual filings.

8.2 Proposed Regulatory Responses

We propose three regulatory interventions:

8.2.1 Venue Diversity Requirements

Proposal: Impose minimum venue diversity requirements on routing, analogous to portfolio diversification requirements for investment funds.

Implementation: Require that no single venue receive more than $X\%$ of a broker's order flow (with X calibrated to prevent excessive concentration while permitting optimization). Our simulation suggests $X \approx 35$–40% would prevent the worst crowding effects while permitting meaningful AI optimization.

Mechanism: A "routing HHI cap" could be implemented similarly to position limits in derivatives markets: soft limits triggering disclosure, hard limits triggering forced diversification.

8.2.2 Correlated Routing Stress Tests

Proposal: Require brokers using AI routing to demonstrate robustness under "correlated routing" scenarios where multiple participants execute similar strategies simultaneously.

Implementation: Similar to bank stress tests (CCAR/DFAST), brokers would submit routing algorithms to a regulator-administered simulation environment that tests behavior when multiple identical strategies interact under stress conditions.

Precedent: This extends the logic of FINRA's existing stress-testing requirements for market makers to the routing dimension.

8.2.3 Enhanced Rule 605/606 Disclosure

Proposal: Amend Rule 605 and 606 to require:

1. Disclosure of whether routing uses AI/ML technology
2. Reporting of routing concentration (HHI) alongside venue destinations
3. Separate execution quality reporting for normal vs. stress periods
4. Disclosure of routing algorithm diversity metrics

8.3 Industry Responses

Beyond regulation, our results suggest several voluntary industry responses:

1. **Anti-correlation mechanisms:** AI routing systems could incorporate "contrarian" components that deliberately diversify away from the predicted consensus routing—analogous to portfolio optimization with a benchmark tracking error constraint.

2. **Venue capacity signals:** Exchanges and ATS could publish real-time capacity utilization metrics, allowing AI systems to avoid congested venues proactively.

3. **Ensemble routing:** Rather than a single AI algorithm, brokers could employ diverse ensembles of routing strategies (Thompson Sampling, UCB, contextual bandits, deep RL) to reduce policy correlation across the industry.

8.4 Welfare Analysis

To quantify the welfare implications, we compare three scenarios:

1. **Status quo (no intervention):** AI adoption continues, crowding externality grows. Net welfare gain relative to all-traditional routing: approximately +\$120M/year during normal times, −\$40M during stress periods. Annual net: approximately +\$80M.

2. **Venue diversity cap ($X = 35\%$):** AI adoption continues but with constrained concentration. Normal-time gains partially preserved (+\$95M), stress losses substantially reduced (−\$12M). Annual net: approximately +\$83M.

3. **Correlated routing stress test:** Brokers internalize crowding risk, develop anti-correlated routing. Normal-time gains mostly preserved (+\$105M), stress losses minimized (−\$8M). Annual net: approximately +\$97M.

These estimates suggest that regulatory intervention can preserve most of the efficiency gains from AI routing while substantially reducing the systemic risks.

9 Conclusion

This paper provides the first systematic analysis of how artificial intelligence transforms the best execution problem in fragmented equity markets. We document a fundamental duality: AI-powered smart order routing achieves significant execution quality improvements during normal market conditions through adaptive venue selection, but creates new systemic fragilities through routing homogeneity and venue crowding during market stress.

Our theoretical framework—modeling AI routing as a multi-agent congestion game—explains why individually optimal learning can produce collectively suboptimal outcomes. When N homogeneous Thompson Sampling routers independently learn over the same environment, they converge to similar policies that concentrate orders at the same venues. This concentration is manageable when market capacity is plentiful but becomes catastrophic when capacity contracts during stress.

Empirically, we find that AI routing reduces execution costs by 26–43 basis points during normal conditions but that this advantage narrows to near zero (or reverses) during stress episodes. The mechanism operates through routing concentration: AI adoption increases venue-level HHI by 20%, with the concentration intensifying during volatility spikes.

Our findings have important implications for financial regulation. Current best execution frameworks—designed for an era of static, diversified routing—are inadequate for evaluating AI-driven routing strategies that create correlated, concentrated execution patterns. We propose venue diversity requirements, correlated routing stress tests, and enhanced disclosure as regulatory responses that can preserve the efficiency benefits of AI routing while mitigating its systemic risks.

Several important questions remain for future research. First, how do AI routing dynamics interact with other forms of algorithmic trading (HFT, execution algorithms)? Second, can mechanism design principles (e.g., congestion pricing at venues) internalize the routing externality without regulatory mandates? Third, how do these dynamics extend to other asset classes (fixed income, FX) where fragmentation is increasing? We leave these questions for future work.

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A Simulation Model Calibration

Our simulation model is calibrated to match the following empirical moments from SEC Rule 605 data (2020–2023 averages):

Target Moments:

Table 9: Calibration Target Moments

Moment	Data	Model
NYSE effective spread (large cap)	1.2 bps	1.2 bps
NASDAQ effective spread	1.4 bps	1.3 bps
IEX effective spread	0.9 bps	0.9 bps
Dark pool effective spread	0.4–0.5 bps	0.4 bps
NYSE fill rate (limit orders)	72–75%	73%
Dark pool fill rate	40–45%	42%
Midpoint match fill rate	25–32%	28%
Normal volatility (VIX 12–18)	—	15% ann.
Stress volatility multiplier	2–3×	2.5×
Spread widening in stress	30–50%	38%

Venue Parameters:

Table 10: Venue Parameter Values

Venue	$\bar{f}$	$\bar{\sigma}$ (cents)	$\bar{\pi}$ (cents)	α	Fee	δ	K
NYSE	0.73	1.2	0.08	0.08	−0.20	0.30	5000
NASDAQ	0.70	1.3	0.12	0.07	−0.25	0.25	5000
IEX	0.55	0.9	0.30	0.03	+0.09	0.12	2000
Dark Pool	0.42	0.4	0.45	0.12	+0.10	0.55	2500
Midpoint	0.28	0.2	0.60	0.15	+0.05	0.70	1500

B Thompson Sampling Algorithm

Algorithm 1 Thompson Sampling for Multi-Venue Routing

Initialize: For each venue $j = 1, \ldots, J$:
  $\alpha_j = 5$, $\beta_j = 3$ (Beta prior for fill probability)
  $\mu_j = 0$, $\tau_j = 100$ (Gaussian prior for reward)

for $t = 1, 2, \ldots, T$ **do**
  Sample fill probabilities: $\tilde{f}_j \sim \text{Beta}(\alpha_j, \beta_j)$ for all j
  Sample rewards: $\tilde{r}_j \sim \mathcal{N}(\mu_j, 1/\sqrt{\tau_j})$ for all j
  Compute scores: $s_j = 0.4 \times \tilde{f}_j + 0.6 \times (\tilde{r}_j \times 1000 + 1)$
  Select venue: $j^* = \arg\max_j s_j$
  Execute order at venue j^*, observe outcome (filled, cost)
  Update posteriors:
  if filled **then**
    $\alpha_{j^*} \leftarrow \alpha_{j^*} + 1$
  else
    $\beta_{j^*} \leftarrow \beta_{j^*} + 1$
  end if
  Update Gaussian: μ_{j^*}, τ_{j^*} using conjugate update
  Apply decay: $\alpha_j \leftarrow \alpha_j \cdot \lambda$, $\beta_j \leftarrow \beta_j \cdot \lambda$ for all j ($\lambda = 0.997$)
  Clip: $\alpha_j = \max(\alpha_j, 1.5)$, $\beta_j = \max(\beta_j, 1.5)$
end for

C Congestion Game Equilibrium Derivation

Proposition 2 (Detailed Proof):

Consider N identical Thompson Sampling routers. Let $\pi_j^{(i)}(t)$ denote router i's posterior mean for venue j at time t. Under the law of large numbers (as the number of observations per venue grows), for any two routers i, i':

$$|\pi_j^{(i)}(t) - \pi_j^{(i')}(t)| \xrightarrow{p} 0 \quad \text{as } t \to \infty \tag{18}$$

This convergence follows because: (1) all routers observe draws from the same underlying venue distributions; (2) with exponential decay ($\lambda < 1$), the posterior is dominated by recent observations; (3) by period t, each venue has been sampled $O(t/J)$ times by each router.

Given identical posteriors, Thompson Sampling generates identical routing probability vectors:

$$\Pr(a_{i,t} = j) = \Pr\left(j = \arg\max_k \tilde{r}_k^{(i)}\right) \to p_j^* \quad \forall\, i \tag{19}$$

where p_j^* is the symmetric equilibrium routing probability for venue j.

The equilibrium routing probabilities $\{p_j^*\}$ satisfy the no-arbitrage condition:

$$r_j(N \cdot p_j^*) = r_k(N \cdot p_k^*) \quad \forall\, j, k \in \text{support} \tag{20}$$

where $r_j(n)$ is the expected reward at venue j when n routers choose it. This is the standard Wardrop equilibrium condition adapted to the routing context.

■

D Additional Robustness Tables

Table 11: Subsample Analysis by Market Cap Quintile

Quintile	Normal Effect	Stress Interaction	Crowding (HHI)
Q1 (Largest)	-0.28^{***}	0.15	$+0.022^{**}$
Q2	-0.35^{***}	0.24^{**}	$+0.038^{***}$
Q3	-0.42^{***}	0.33^{***}	$+0.051^{***}$
Q4	-0.48^{***}	0.39^{***}	$+0.062^{***}$
Q5 (Smallest)	-0.52^{***}	0.45^{***}	$+0.071^{***}$

Notes: The crowding externality is monotonically stronger for less liquid securities, where venue capacity constraints bind more frequently.

Table 12: Placebo Tests — Random Adoption Dates

Percentile	Placebo Normal Effect	Placebo Stress Interaction
5th	-0.18	-0.14
25th	-0.06	-0.04
50th (median)	0.01	0.02
75th	0.07	0.08
95th	0.16	0.15
True estimate	$\mathbf{-0.38}$	**0.31**
p-value (one-sided)	< 0.01	< 0.01

Notes: Distribution based on 1,000 random permutations of adoption dates. True estimates fall well outside the placebo distribution.

E Variable Definitions

Table 13: Variable Definitions

Variable	Definition	Source
Effective Spread	$2 \times \lvert\text{Trade Price} - \text{Midpoint}\rvert/\text{Midpoint} \times 10{,}000$	Rule 605
Price Improvement	$⊮\{\text{Trade Price better than NBBO at arrival}\}$	Rule 605
Fill Rate	Orders Fully Executed / Orders Received	Rule 605
Execution Speed	Median ms from order receipt to execution	Rule 605
Routing HHI	$\sum(\text{venue share})^2$ across venues receiving orders	Rule 606
AI Adoption	$⊮\{\text{Broker disclosed AI routing in quarter } t\}$	10-K, Press
HighVol	$⊮\{\text{VIX} >$ 75th percentile trailing 252 days$\}$	CBOE
AIExposure	$\sum$ broker market share $\times$ adoption indicator	Rule 606
Total Cost	Effective spread + fees − rebates	Rule 605 + Fee schedules

End of Paper